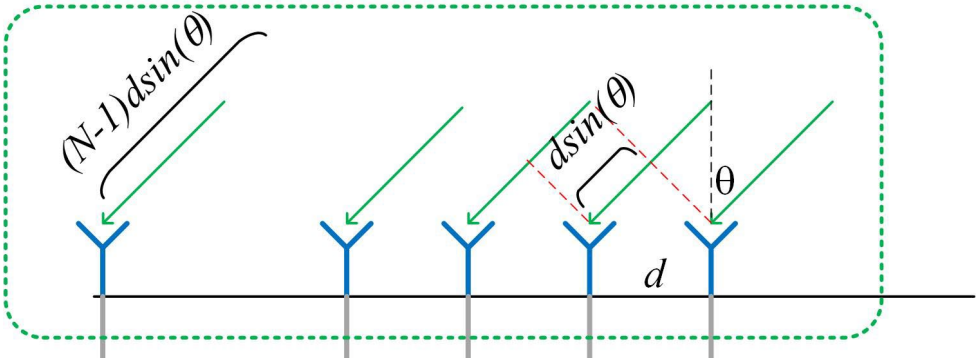
Digital Communication IC Design

Homework #4

(Due on 2026/11/18)

- Please upload your codes and reports (with simulation results and required explanations) to the NTUCool.
- Please disclose how you use AI tools and its version.

Total points: 145 points



1. Let

$$\mathbf{h} = \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \alpha_l \, \boldsymbol{a}^{\boldsymbol{ULA}}(\theta_{r,l})$$

and

$$\boldsymbol{a}^{\boldsymbol{ULA}}(\theta) = \frac{1}{\sqrt{N_r}} [1 \quad e^{-jkdsin(\theta)} \quad \dots \quad e^{-j(N-1)kdsin(\theta)}]^T.$$

Assume $kd = \pi$ and 64($= N_r$) receiving antenna elements are used in the uniform linear antenna array. The transmitter has one antenna. Given $L = 2$,

(a) please generate your own $\theta_{r,1}$ and $\theta_{r,2}$ such that $-\frac{\pi}{3} < \theta_{r,1}, \theta_{r,2} < \frac{\pi}{3}$. Report them in both radians and degrees. Set $\theta_{t,1} = \pi/7$. Please generate your channel matrix $\mathbf{h}$ with AoA $\theta_{r,1}$ and $\theta_{r,2}$, which is a vector of dimension 64×1. Let α_1 and α_2 be the complex channel coefficients with $|\alpha_1| > 0.5 > |\alpha_2|$. Please generate them randomly and report their magnitudes and phases. (5%)

(b) Now let's check the effect of a receiver performing beamforming. Given that

$$\boldsymbol{c}(m) = \frac{1}{\sqrt{N_r}} [1 \quad e^{-j2\pi m/N} \quad \dots \quad e^{-j2\pi(N-1)m/N}]^T$$

is used as the phase shift for combiner at the receiver and $m = -32, -31, \dots, 31$. First, use your last digit of student ID as m_1. Then, draw the beam pattern associated with m_1. Namely, fix $\boldsymbol{c}(m_1)$ and scan $\left|\boldsymbol{c}(m_1)^{\boldsymbol{H}} \boldsymbol{a}^{\boldsymbol{ULA}}(\theta)\right|$ for $-\pi/2 \le \theta \le \pi/2$ in polar coordinate. (10%)

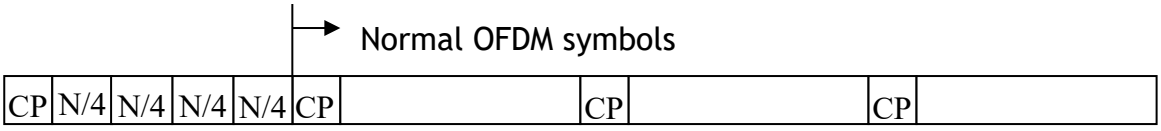
(c) Please check the magnitude of the combination result, $|\boldsymbol{c}(m)^{\boldsymbol{H}} \mathbf{h}|$ versus m. Draw the figure with x-axis indicating the value of m and y-axis is the magnitude of $|\boldsymbol{c}(m)^{\boldsymbol{H}} \mathbf{h}|$ for $m = -32, -31, \dots, 31$. (10%)

(d) If $\hat{m} = arg\ max\ |\boldsymbol{c}(m)^{\boldsymbol{H}} \mathbf{H_1}|$, then calculate the nominal pointing angle $\varphi(\hat{m})$ associated with the **beamforming vector** $\boldsymbol{c}(\hat{m})$ (5%). Please check the differences of

$\left|\varphi(\hat{m}) - \theta_{r,1}\right|$ and $\left|\varphi(\hat{m}) - \theta_{r,2}\right|$. (5%)

(e) Explain whether the selected beam is closer to the stronger path, the weaker path, or neither. Support the explanation using the path magnitudes, path phases, angular separation, and the values. (5%)

2. Please download "hw4.mat" from the NTUCool. Use "load hw4.mat" to retrieve the original data. It contains one variable "OFDMTx", which are the complex time-domain discrete signals generated by OFDM baseband transmitter. There are 32 samples in cyclic prefix ($N_g = 32$) and the FFT size is 128 ($N = 128$). The first OFDM symbol is a preamble which has identical two repetitions as below, and then three normal OFDM symbols follows the preamble. Assume the received signal z_m equal to the transmitted signal. If $m \leq 0$, $z_m = 0$.



Symbol 1 Symbol 2 Symbol 3

(a) Please draw magnitude of the delay and correlate result $\Phi_{DC}(m)$ by setting $R = 64$ and $L = 64$, for $m = 1,\ldots,640$ where

$$\Phi_{DC}(m) = \left|\sum_{r=0}^{R-1} z_{m-r} z_{m-r-L}^*\right|. \text{ (10%)}$$

(Mark the X and Y value of the peak in the figure. The first index in the figure is 1.)

(b) Consider only the first symbol for $1 \leq m \leq 160$. If a decision is made according to $\hat{m}_{DC} = \arg\max \Phi_{DC}(m)$, what's the detected timing in the first symbol? (5%) Whether a plateau will be seen or not in the waveform of $\Phi_{DC}(m)$ using the setting in Q2(a) ? Why? (5%)

(c) Discard the first symbol. Now, the signal length is 480 sample. Please draw the result $\Phi_{NM}(m)$ and $\Phi_{DC}(m)$ by setting $R = 32$ and $L = 128$ for $m = 1,\ldots,480$. If a decision is made according to $\hat{m}_{DC} = \arg\max \Phi_{DC}(m)$ and $\hat{m}_{NM} = \arg\max \Phi_{NM}(m)$ every symbol, what are the results? Please explain the decision difference between $\Phi_{NM}(m)$ and $\Phi_{DC}(m)$ (15%)

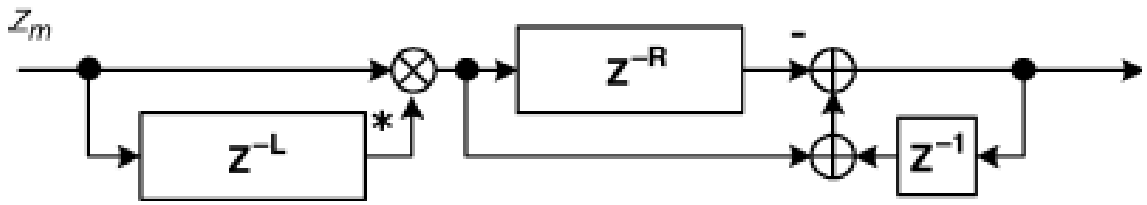
$$\Phi_{NM}(m) = \frac{\left|\sum_{r=0}^{R-1} z_{m-r} z_{m-r-L}^*\right|^2}{\left(\sum_{r=0}^{R-1} |z_{m-r}|^2\right)^2}$$

(d) Now, add noise by yourself. The SNR is 12dB. Please use normalized metric $\Phi_{NM}(m)$ to perform symbol timing detection again. Report the detected result $\hat{m}_{NM} = \arg\max \Phi_{NM}(m)$ every symbol and compare the results to the answer in Q2(c). Please explain the difference if any. (10%)

3. Now, we aim to implement $\Phi_{NM}(m)$. A constant threshold η is set to activate the peak searcher. It means the local maximum will be searched only when the correlation output is larger than η, namely

$$\Phi_{NM}(m) = \left|\sum_{r=0}^{R-1} z_{m-r} z_{m-r-L}^*\right| > \eta$$

The delay correlation is implemented for the numerator as in the following.



(a) Decide your own setting of η from the given signals and the signals with SNR equal to 12dB. Please report the setting and explain the reason. (5%)

(b) Assume that the complex input signal enters one by one sequentially. Draw the block diagram of your design for the threshold-judgement module, divider, and the peak searcher together with the delay correlator. Connect all these modules together in one figure. (10%)

(c) Please quantize your input signal z_m by 16 bits. To avoid overflow, please explain the wordlength setting of your accumulator and your divider. (5%)

(d) Implement your design by hardware description language (HDL) (20%). Explain the latency of your local peak searcher (5%). Show the behavior simulation results (timing diagram) of your symbol boundary detector (10%). Please compare your implementation results with your simulation result in Q3(c) and provide your comment. (5%).